Unit: 6 Unit 4: Absolute Value
Topic: Solving Absolute Value Equations

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. Solve the equation $|x| = 5$
(A) $x = 5$
(B) $x = -5$
(C) $x = 5$ or $x = -5$
(D) $x = 0$
(E) $x = 10$

Q2. Solve the equation $|2x| = 8$
(A) $x = 2$
(B) $x = -2$
(C) $x = 2$ or $x = -2$
(D) $x = 4$
(E) $x = -4$

Q3. Solve the equation $|x - 3| = 2$
(A) $x = 1$
(B) $x = 5$
(C) $x = 1$ or $x = 5$
(D) $x = 3$
(E) $x = 0$

Q4. Solve the equation $|x + 2| = 4$
(A) $x = 2$
(B) $x = -6$
(C) $x = 2$ or $x = -6$
(D) $x = 0$
(E) $x = -2$

Q5. Solve the equation $|3x - 1| = 5$
(A) $x = 2$
(B) $x = -2$
(C) $x = 2$ or $x = -\frac{2}{3}$
(D) $x = \frac{2}{3}$
(E) $x = -\frac{2}{3}$

Q6. Solve the equation $|x^2 - 4| = 8$
(A) $x = 2$
(B) $x = -2$
(C) $x = 2$ or $x = -2$ or $x = 4$ or $x = -4$
(D) $x = 0$
(E) $x = 4$ or $x = -4$

Q7. Solve the equation $|2x + 5| = |x - 2|$
(A) $x = -1$
(B) $x = 1$
(C) $x = -1$ or $x = \frac{7}{3}$
(D) $x = \frac{7}{3}$
(E) $x = -\frac{7}{3}$

Q8. Solve the equation $|x - 2| = |3 - x|$
(A) $x = 1$
(B) $x = 2$
(C) $x = 2.5$
(D) $x = 1$ or $x = 2$
(E) $x = 2.5$ or $x = 0$

Open Ended Questions (FRQ)

Q9. Solve the equation $|x^2 + 2x - 6| = 2$ and check for extraneous solutions

Q10. Solve the equation $|3x - 1| = |2x + 1|$ and check for extraneous solutions

Chef's Tips

- Check for extraneous solutions
- Use the definition of absolute value to solve equations
- Consider all possible cases when solving absolute value equations